

Template Checklist/Rubric: This is how I grade you: Each implementation item will garner 5 points for a total of 100.

1. Completeness of Work

Constructs		Parser	Semantics	Interpreter
Headers/Comments (not required)				
Variable Declaration				
Pointers				
Structure				
Constant Declaration				
Assignment				
Math Expr	Simple			
	Complex (includes arrays, function calls)			
Conditional Statements				
Iterative Statements	Event-controlled loop			
	Count-controlled loop			
	Repeat-Until/Do-while			
Boolean Expressions	Simple			
	Complex/Simple Logical			
	Multiple/Complex Logical			
Input Statement				
Output Statement				
Functions/Methods	Declaration and Calling			
	Variable Scope			
	Function Calling Another Function			
	Call Multiple Times			
	Recursion			
Nested Statements				

2. Error Messaging:
 - a. Accuracy of Message
 - b. Informativeness
3. Error Recovery
4. Robustness
5. Flexibility and Robustness of Expressions

Following the Expression Hierarchy:

 - a) simple_math:

e.g. $x = id + num;$

 - i. $x = id * id;$
 - ii. $x = num / num;$
 - b) complex_math: nested simple_math, multiple simple math + function calls + arrays

e.g. $x = func1(23,4,5) + 34/234*z32;$
 - c) simple boolean: rel_op : <, >, <=, >=, !

e.g. (simple_math rel_op simple_math)

 - i. (complex_math rel_op complex_math)
 - d) complex_boolean: logical_op: &&, ||

e.g. (simple_boolean logical_op simple_boolean)
 - e) Complex_Logical: nested complex_boolean

(complex_math rel_op complex math logical_op !(complex math) ...)
6. Efficiency
7. Other Comments
8. Heap Simulation

Semantic Errors

1. Undeclared variable
2. Type Mismatch
3. Multiply-defined variable
4. Constant Reassignment
5. Cardinality/Ordinality: function parameters

	10	7	4	0
Execution Modes	Code can be executed all at once or line per line.	Code can be executed line per line but details cannot be seen (symbol table, call stack)	Code cannot be executed line per line.	Code cannot be executed at all.
Symbol Table and Call Stack	Symbol table and call stack show the correct information while the programming is running.	Symbol table and call stack are present but some information shown are inaccurate / inconsistent.	Only one of them is present.	Symbol table and call stack are not shown at all.
Error Messages	Errors are correctly detected and informative messages are generated.	Errors are correctly detected but the messages are not helpful.	Errors detection is inaccurate or lacking.	There is no error detection at all.